Akash Dubey

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Founding & Product Engineering

Scarlet Sync New Brunswick, NJ
Founder *Jan 2025 – Present*

- Built and shipped a modern Rutgers academic planning platform from 0 to 1 to thousands of users for degree, schedule, and class planning plus campus discovery, utilizing AI-assisted parsing of unstructured data from legacy university services.
- Architected an end-to-end data ingestion pipeline using Python and LLMs to reverse-engineer legacy data, indexing over 2,500 classes and 500+ degree programs for real-time querying.
- Developed a production AI assistant with streaming tool-calling and grounded retrieval over schedules, degree plans, and profile context, enabling automated schedule building and personalized academic planning across Web, ChatGPT, Claude, and iOS.

Lykke San Francisco, CA
Founding Software Engineer *Sep 2025 – Present*

- Built core AI workflows for an AI study platform used to ingest Canvas LMS courses, assignments, announcements, notes, and uploaded documents into class-scoped knowledge bases for contextual chat and study generation.
- Engineered adaptive RAG pipelines across MongoDB, Zilliz/Milvus, S3, Gemini Files API, and SSE streaming, including source inference, citation events, token-budgeting for large course materials, and fallback paths for extracted text vs. full-file reasoning.
- Created a distributed course-wiki generation engine that crawls educational sources, synthesizes structured sections, generates interactive artifacts, embeds wiki content for retrieval, and runs tenant-isolated background builds via SQS, MongoDB leases, Redis rate limiting, and AWS ECS.

Samaritan Scout Cranford, NJ
Full Stack Engineer Intern *May 2023 – Aug 2025*

- Developed a full-stack search engine using React, TypeScript, and PostgreSQL to index 35,000+ volunteer opportunities, implementing semantic search algorithms with hallucination filters to ensure accurate matching.
- Developed an agentic scraping system leveraging OpenAI and Gemini structured outputs to autonomously extract data from over 100,000 non-profit websites, reducing manual data entry time by over 90%.

New York Life Insurance New York, NY
Software Engineer Intern, TDAV Team *May – Aug 2026*

- Created an AI text and voice agent for processing insurance claims for clients and 12,000 Customer Service Representatives using AWS Infrastructure with access to NYL services, knowledge, and forms.
- Constructed a company wiki/knowledge base from 13,000 documents compiled from Sharepoint, Public Site, Confluence, Jira, and the Intranet bringing business logic into one place for AI and human users.

Selected Technical Projects

Knowledge Base and AI Agent Workflow Manager | *Data Ontology, Workflows, Coding Agents*

- Built KRAIL (pypi), a local-first, repo-backed knowledge runtime that structures raw notes into ontology-aware topic pages, markdown graphs, source ledgers, claim records, verification runs, and artifact lineage for evidence-grounded agent memory.
- Implemented MCP/CLI and agent infrastructure for auditable coding-agent workflows (for software factories, auto-research, and company brain use cases), including local runner dispatch, dry-run work orders, session logs, role prompts/checklists, workflow validation, and hybrid keyword-graph-vector retrieval.

Local Coworker: Native macOS Computer-Use Agent | *Swift, MCP, Gemini Vision* (private code)

- Built a native desktop agent capable of autonomous computer use by integrating Gemini Vision with macOS Accessibility APIs, AppleScript, and JXA.
- Implemented the Model Context Protocol (MCP) to orchestrate specialized sub-agents (UI Interaction, Shell, App Research) for planning and executing complex multi-app workflows.

SLM Optimization & Post-Training (GSM8k) | *PyTorch, HPC*

- Fine-tuned small language models (SLMs) on the GSM8k math reasoning dataset to study the effects of model size and architecture on math reasoning, deploying on the Rutgers HPC (Amarel) cluster with quantization and structural pruning to maximize inference throughput on constrained compute.

Research

Rutgers Economics Labs New Brunswick, NJ
President (formerly Research Director, Team Lead) *Oct 2024 – Present*

- Architected RAIL (Rutgers Agentic Intelligence Labs), an AI automated economic research pipeline to collect, organize, analyze, and synthesize economic data using data ontologies and coding agent workflows.
- Lead quantitative research for partners including NJDOL, NJDEP, NJBPU, Federal HUD, and the U.S. Census Bureau; applied econometric models in Python/R to analyze large datasets and project economic trends.

Algorithmic Robotics and Control Lab in Rutgers's Dept. of CS New Brunswick, NJ
Research Assistant (Advisor: Prof. Jingjin Yu) *Jan 2026 – Present*

- Implemented and validated CUDA versions of pRRTC and FCIT*, parallelizing tree/graph search and collision checking for high-dimensional robot motion planning across 500 benchmark problems.

Kwan Lab, Rutgers Cancer Institute of New Jersey New Brunswick, NJ
Computational Biology Research Assistant (Advisor: Prof. Kelvin Kwan) *Jan 2026 – Present*

- Built an AlphaGenome-based in silico deletion screen (72 perturbation conditions) and a 3D chromatin polymer simulator validated with 70+ automated tests to quantify TAD-boundary sensitivity from Hi-C contact matrices.

Education

Rutgers University Honors College, School of Arts and Sciences New Brunswick, NJ
Bachelor of Science in Computer Science and Mathematics *Anticipated May 2027*

- **GPA**: 3.95/4.0 | SAT: 1580
- **Relevant Coursework**: Tensor Networks, Systems Programming, Stochastic Processes, Computer Architecture, Algorithms, Data Structures, Honors Calculus III & IV, Math Theory of Probability.

Awards & Skills

Awards: 8090 AI Top Coder (5th), Rutgers Shark Tank (2nd and 3rd), ASA Fall Data Challenge (Top 3), NSF I-Corps.
Languages & Frameworks: Python, TypeScript, React, SQL, C++, Java, R, Bash, Git
AI & Product: LLMs, RAG, Agents, Tool-Calling, MCP, Transformers, Generative AI, PyTorch
Infrastructure: AWS, PostgreSQL, MongoDB, Redis, S3, ECS/SQS, Vector DBs (Milvus/Zilliz), HPC